Paleomagnetism of some late Precambrian and lower Paleozoic sediments from L'Adrar de Mauritanie, West Africa

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Palaeomagnetic results are reported from three formations from the late Precambrian to lower Paleozoic sedimentary sequence of L'Adrar de Mauritanie, on the West African Craton. Each of the formations recorded complex magnetizations, the sequence of which could only be resolved after detailed thermal and chemical cleaning. Most of these complexities have arisen from post depositional hematite coating on both magnetic and non-magnetic grains. Formation I_2 with a probable age of 1020 Ma records an early, presumably original remanence directed along $D = 034°$, $I = +60°$ ($\alpha_{95} = 6°$) and a later remanence $D = 336°$, $I = +37°$ ($\alpha_{95} = 11°$). Specimens from formations CO_8 and CO_{10} (Cambro–Ordovician boundary) record a common late phase remanence at $D = 128°$, $I = +24°$ ($\alpha_{95} = 8°$). The CO_{10} early phase remanence is $D = 033°$, $I = +61°$ ($\alpha_{95} = 7°$), whereas the poorly isolated early phase remanence of CO_8 approximates $D = 060°$, $I = +51°$ ($\alpha_{95} = 20°$).

On rapporte les résultats paléomagnétiques de trois formations de la séquence sédimentaire de l'Adrar de Mauritanie, dans le craton ouest-africain, datant de la fin du Précambrien au Paléozoïque inférieur. Chacune des formations présente des aimantations complexes et pour établir la séquence, il faut faire subir un traitement thermique et chimique élaboré aux échantillons. Une grande partie de cette complexité est attribuable à l'enrobage par de l'hématite de grains magnétiques et non magnétiques après le dépôt. La formation I_2, datant probablement de 1020 Ma, a enregistré une rémanence précoce, probablement originelle, dirigée selon $D = 034°$, $I = +60°$ ($\alpha_{95} = 6°$) et une rémanence plus récente selon $D = 336°$, $I = +37°$ ($\alpha_{95} = 11°$). Des spécimens de la formation CO_8 et CO_{10} (à la limite du Cambro-Ordovicien) ont enregistré une phase tardive commune de rémanence selon $D = 128°$, $I = +24°$ ($\alpha_{95} = 8°$). La phase initiale de rémanence dans la formation CO_{10} est selon $D = 033°$, $I = +61°$ ($\alpha_{95} = 7°$), alors que la phase de rémanence faiblement isolée de la formation CO_8 est approximativement selon $D = 060°$, $I = +51°$ ($\alpha_{95} = 20°$).

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Introduction

Late Precambrian sedimentary sequences that overlie many of the cratonic regions of Africa have been partially incorporated in the anastomosing Pan-African mobile belts (Kennedy 1964; Clifford 1968, 1970). Whether these belts are the signatures of ancient plate boundaries (Burke and Dewey 1972) or regions of thermal and tectonic reactivation (Kennedy 1964; Rocci 1964, 1965) is open to discussion (Shackleton 1976). Because the African late Precambrian paleomagnetic record is presently deficient in temporal sequence, discussion of the palaeomagnetic evidence for, or against, Pan-African oceanic development is speculative. Previous attempts to show the African cratons have maintained their present relative positions since the Archean have been based largely on the ability to construct a single smooth pole path (Briden 1976; Piper 1976). In this paper preliminary results are reported for the first time from the late Precambrian clastic sequence in 'L'Adrar de Mauritanie' region of the West African Craton (Fig. 1). Because only a limited number of sites were occupied, only the poles from formation I_2 could be used for regional tectonic analyses.

Formations Sampled

The sedimentary sequence from 'L'Adrar de Mauritanie', West Africa, which extends from late Precambrian to Upper Devonian, lies unconformably upon basement gneisses (Fig. 2) dated at ap-

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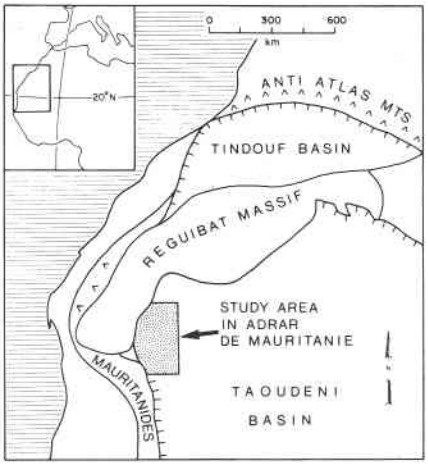


FIG. 1. General location map and schematic structural map of West Africa showing the study area in the Adrar de Mauritanie (after Trompette 1973).

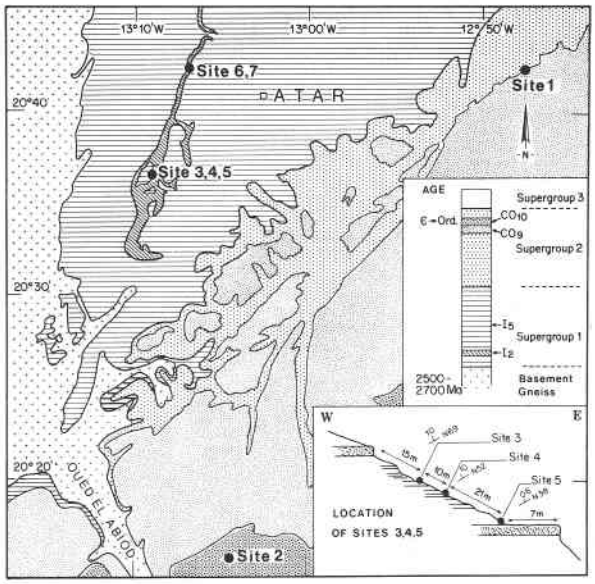


FIG. 2. Locality map showing the distribution of Series 1 and 2, and the individual sampling localities. Inserts show (a) the location of the sampled formations within the three series, and (b) the relative stratigraphic positions of sites 3–5.

proximately 2500 Ma (Bonhomme 1962; Giraudon and Vachette 1964). On the basis of angular unconformities, disconformities and distinctive stratigraphic horizons, Trompette (1973) has subdivided these pre-Devonian rocks into three supergroups. Further subdivision of each supergroup into formations on the basis of lithology is less important. During sedimentation of the three supergroups tectonic disturbance of the craton was weak but widespread; climatic phenomena exercised a dominant control on lithology. High latitude sedimentation is represented by two extensive glacial deposits, whereas low latitude deposition is represented by stromatolite banks interbedded with evaporites. Subsequent to deposition, there has been little or no tectonic or metamorphic disturbance (Clauer 1976).

As these subdivisions are lithostratigraphic, with the first good stratigraphic fossils occurring in the Silurian, the chronology of these pre-Silurian rocks is hypothetical. Age control on the units sampled for paleomagnetism is not precise. A Rb/Sr date of 1019 ± 32 Ma[2], reported from illites of the lowest Groupe de Char (I_2 of Fig. 7), provides an estimate for the age of initiation of sedimentation on the craton (Clauer and Bonhomme 1971). Age control on the second group sampled, le Groupe des Plateaux d'Oujeft (CO_9–CO_{13} of Fig. 7), is less precise. Two inarticulate brachiopod faunas suggest deposition around the Cambro–Ordovician boundary.

Experimental Procedure and Data Treatment

Up to 10 cores, referred to as samples, were field drilled at each of the seven sites, one site each from formations CO_8 and CO_{10}, and five sites from formation I_2. A maximum of three specimens were obtained from each core. At each site sampling was restricted to a specific stratigraphic horizon having a measured bedding tilt. Core orientation was by sun and (or) magnetic compass. All remanence measurements were made on a PAR-SM1 spinner magnetometer at the University of Western Ontario. *All* specimens were systematically investigated by one or two stepwise treatment techniques. Thermal demagnetization, in eight steps up to 690°C, was performed in a furnace designed by the authors. In this the specimens were rotated about a horizontal axis oriented east–west, while being heated and cooled in a mumetal box. Alternating field demagnetization, in 10 steps up to the maximum available of 1000 Oe, was achieved in a three axis tumbler. Chemical cleaning, in five steps up to 2140 h was aided by specimen serration. A total of eight cuts were made into each core by a $\frac{3}{8}$ in. (9.5 mm) thin wall drill. The cuts were made in four opposing pairs to annul any remanence contamination which may have arisen from the drilling process. To optimize the total surface area exposed, together with the least reduction in volume, the cuts were made so as to leave the central core attached (Fig. 5).

Each specimen carried more than one remanence. Vector analysis of the thermally and chemically cleaned natural remanent magnetization (NRM) remanence data, employing the method of

[2]Rb/Sr decay constant $\lambda = 1.39$.

TABLE 1. Remanence directions for formations CO_8 and CO_{10}*

Formation (location)	N(n)†	R	k	α_{95}	D (tilt corrected)	I (tilt corrected)	Lat. (palaepole) deg	Long. (palaepole) deg
Site 1								
CO_8-I	4(4)	3.821	17	23	060	51	36 N	052 E
CO_8-S	7(7)	6.975	241	4	126	20	28 S	052 E
Site 2								
CO_{10}-I	7(22)	6.931	87	7	033	61	55 N	030 E
CO_{10}-S	7(22)	6.907	65	8	128	27	28 S	048 E
CO_8-S + CO_{10}-S	14	13.853	89	4	127	23	28 S	050 E

*Location 20 N, 347 E.
†*N* represents the number of core mean directions; *n*, the number of specimen mean directions.

Roy and Park (1974), usually allowed the direction of each magnetization phase to be determined. Individual specimen remanence directions could then be identified by a number of methods; direct measurement or vector substraction during either thermal or chemical cleaning.

The results are presented as they relate to each group sampled, firstly for formations CO_8 (site 1) and CO_{10} (site 2) of the Groupe des Plateaux d'Oujeft, and then for the five sites (3–7) of formation I_2 in Groupe de Char.

Results

Supergroup 2

The stratigraphical and topographical locations of sites 1 and 2 are summarized in Figs. 1 and 2. Specimens from site 1, CO_8 from the Groupe de la Falaise d'Atar are distinctly different than those from site 2, CO_{10} of the Groupe des Plateaux d'Oujeft. The compositional and textural differences of the specimens are amplified by analysis of their remanent magnetism.

Analysis of the thermal and chemical treatment data indicate early phase (I) magnetizations are carried by detrital particles (of both hematite and magnetite), whereas the later (S) magnetizations reside in the red hematite pigment. The early phase (I) remanence directions carried by formation CO_{10} ($D = 033°$, $I = +61°$, $\alpha_{95} = 7°$) and formation CO_8 ($D = 060°$, $I = +51°$, $\alpha_{95} = 23°$) have some directional similarity (Table 1). The lack of statistical identity may be in part due to poor isolation of the CO_8 initial remanence as discussed below. Conversely the late phase (S) magnetizations of CO_{10} ($D = 128°$, $I = +24°$, $\alpha_{95} = 8°$) and CO_8 ($D = 126°$, $I = +20°$, $\alpha_{95} = 4°$) are statistically identical, and therefore probably relate to the same post-depositional event (Table 1). These results must be considered preliminary, because of limited sampling.

CO_8

Remanence treatment techniques indicate the presence of two remanence directions carried by two magnetizations of very different physical and chemical properties. The late phase (S) direction is carried by fine grained impure hematite with a blocking temperature around 550°C, which can be partially leached after 2000 h of acid treatment. The early phase (I) direction is carried by coarse-grained hematite with a broad range of blocking temperatures, which is partially revealed after extensive acid treatment.

Silt to clay particles dominate the grain size distribution of the CO_8 specimens. In outcrop the beds are dark red, well laminated, and exhibit small scale flame structures. Hematite, which occurs almost exclusively in the red pigment form, dominates the remanence of the CO_8 specimens. Their NRM directions are closely clustered around the secondary remanence direction (S) as is shown for samples 1–31, 1–51 in Fig. 3. Thermal demagnetization up to 500–550°C produces no significant change of magnetic intensity or remanence direction. However, after 570–600°C, the direction suddenly migrates to the I remanence position and the intensity drops by over 70%. Further thermal treatment produced a slight intensity decay, with no direction change until all remanence is eradicated at 700°C. Chemical treatment produced a progressive decay of remanent intensity concomitant with a gradual shift from the secondary (S) to the initial (I) position. Even after more than 2140 h of chemical leaching the remanence direction of some specimens had still not achieved the I position. These specimens retained a core of unleached hematite pigment, but further leaching was not possible because at this point the cores had become exceedingly fragile, and began to disintegrate.

These argillites are relatively impermeable and even with specimen serration chemical leaching was very slow owing to the compact nature of the sediment. Each successive acid immersion removed a certain portion of the cementing hematite pigment, thereby altering the lithologic characteris-

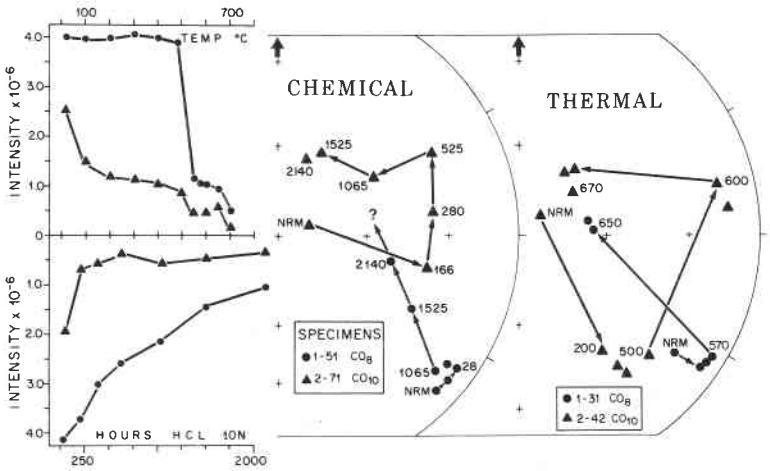


FIG. 3. Results obtained from the thermal and chemical treatment of samples from formations CO_8 (site 1) and CO_{10} (site 2) showing their different cleaning characteristics. Both exhibit a progression toward the more northerly steeply inclined direction after extensive treatment. Chemical leaching of the CO_8 specimen did not remove all the red pigment, although progressing toward the site CO_{10} specimen stable end-point this is not achieved even after 2140 h in acid. Throughout, solid symbols conventionally represent downward directed remanence.

tics of the leached zone to a more permeable and friable state. In this mode, acid attack of specularite grains in the now more permeable zone proceeds along with further hematite pigment removal, and hence enlargement of the increased permeability zone. Ideally, chemical leaching preferentially dissolves the hematite pigment and its associated remanence, leaving untouched the presumably detrital specularite. In low permeability rocks separation of multiple remanence directions by acid leaching is often difficult, since both remanence carrying phases are being attacked concurrently. In this instance the chemical degradation rate of the pigment exceeded that of the specularite, and the specularite remanence should have been well isolated by further chemical leaching, but the cores became too fragile to allow further treatment.

CO_{10}

Detailed chemical and thermal cleaning of the CO_{10} specimens revealed three magnetizations: a recent one (R) carried by goethite, a diagenetic one (S) carried by hematite pigment, and a presumably initial depositional one carried by specularite (I).

Lithologically, the CO_{10} formation is quite different, consisting of cross-bedded, feldspathic medium grained sandstones. In colour the beds are mauve, red hematite pigment occurring only as a minor grain coating. Post-diagenetic percolation of water through this permeable sandstone has produced iron-rich liesegang banding sub-parallel to the present day surface. Unlike the CO_8 specimens, the NRM directions of the CO_{10} specimens prior to cleaning were steeply inclined to the northeast (Fig. 3). Upon commencement of thermal and chemical treatment, the magnetization of the specimens exhibited a rapid direction change toward a much shallower inclination. The new direction revealed by thermal cleaning is identical to the S magnetization found in the CO_8 specimens. Associated with this rapid direction change, the specimens show a significant intensity decrease; less than 50% of the NRM intensity remains after 200°C or 166 h in acid. The vector removed during this direction shift, as calculated by the vector subtraction of D, I (250°C) from D, I (NRM), closely approximates to the I remanence direction derived from extended thermal and chemical cleaning, and also crudely approximates the local geomagnetic field direction of the sampling locality.

This removed remanence could then either record an early depositional magnetization or a recent chemical magnetization. Detailed analysis of the initial steps of chemical cleaning are less informative. After the first cleaning step (166 h), the new direction does not agree with the commonly found S magnetization direction, and consequently the removed vector, although northerly and steeply inclined, does not as closely approximate to the I remanence direction. The sandy CO_{10} specimens are quite permeable, total acid penetration was probably achieved during each period of immersion, in which case acid attack would be rapid. Because we have no information between 0 and 166 h, it is impossible to decipher just how rapidly this magnetization was chemically degraded and therefore the detailed direction changes associated with

TABLE 2. Remanence directions for formation I_2*

Site	N†	R	k	α_{95}	D (dip corrected)	I (dip corrected)	Lat. (paleopole) deg	Long. (paleopole) deg
3-I	7	6.893	56	8	038	64	50 N	029 E
4-I	5	4.953	53	10	028	58	60 N	033 E
5-I	4	3.879	25	18	026	54	63 N	040 E
6-I	3	2.954	48	18	047	60	45 N	039 E
7-I	4	3.979	142	8	042	65	47 N	029 E
I mean	5	4.975	160	6	034	61	54 N	032 E
3-S	7	6.874	48	9	334	40	66 N	267 E
4-S	4	3.975	125	8	329	42	61 N	271 E
5-S	4	3.899	30	17	334	43	66 N	273 E
6-S	2				328	53	59 N	289 E
7-S	8	7.512	14	15	342	18	69 N	129 E
S mean	5	4.888	36	11	336	37	67 N	258 E

*Location 20.3 N, 346.9 E.
†*N* in site mean calculations is the number of sample mean observations, and in overall calculations the number of site mean observations.

this degradation are unknown (Fig. 3). Physically this remanence is characterized by low acid resistance (166 h) and low blocking temperature (200°C). Unless its carrier is in much greater abundance than that of the remaining remanence phases, it must also have a much higher specific magnetization. Coarse grained titanium-rich magnetite (of detrital origin) and goethite (possibly of recent origin) are the two most likely possibilities. The latter interpretation is favoured for the following reasons. Firstly, no coarse magnetite was observed in any of the polished thin sections (Fig. 5). Secondly, the iron-rich liesegang banding, which was clearly visible in the original specimens, disappeared after acid leaching. Thirdly, during thermal treatment disappearance of the banding was associated with a significant reddening of the specimens, possibly due to the dehydration of goethite to hematite (Strangway *et al.* 1968).

Further chemical and thermal treatment of the CO_{10} specimens isolated two other magnetizations. Thermally, the common S direction persisted until approximately 500–550°C, when the remanence direction migrated towards the I remanence direction (Fig. 3), as occurred in the CO_8 specimens. The difference in magnitude of the intensity drop associated with this direction change reflects the difference in red hematite pigment content between the CO_8 and CO_{10} specimens. Further chemical leaching produced a slow intensity degradation and direction change until after 1500 h the initial remanence direction was achieved.

Supergroup 1

Le Groupe de Char—Formation I_2 (1019 + 32 Ma)

The stratigraphical and topographical locations of sites 3–7 are summarized in Figs. 1 and 2. All sites were located in Formation I_2 of the Groupe de Char. Red siltstone and shale, interbedded with dolomite, constituted over 90% of the outcrop. At sites 6 and 7 greenish spherical reduction spots, roughly 1–2 cm across, were very common. Samples were collected over a total stratigraphic range of 40 m and horizontal extent of 20 km. Thermal

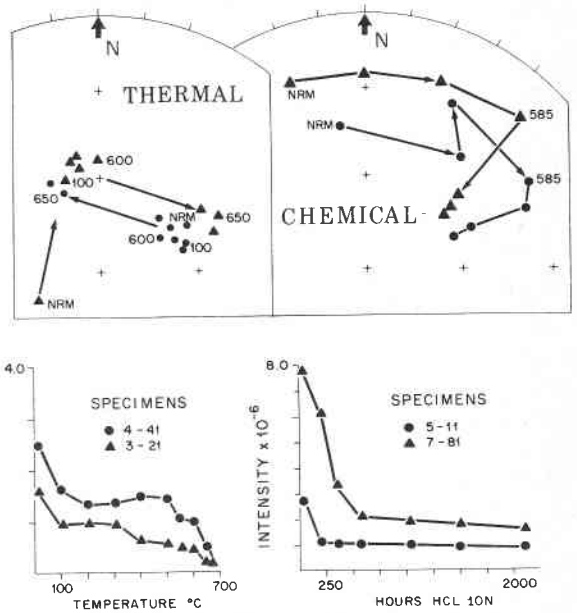


FIG. 4. Results obtained from the thermal and chemical treatment of samples from formation I_2. The uniformity of the lithology is reflected by the general similarity of the direction and intensity changes upon cleaning. Although prolonged thermal cleaning may reveal either of the stable directions, acid cleaning preferentially reveals the steeply inclined northeasterly direction.

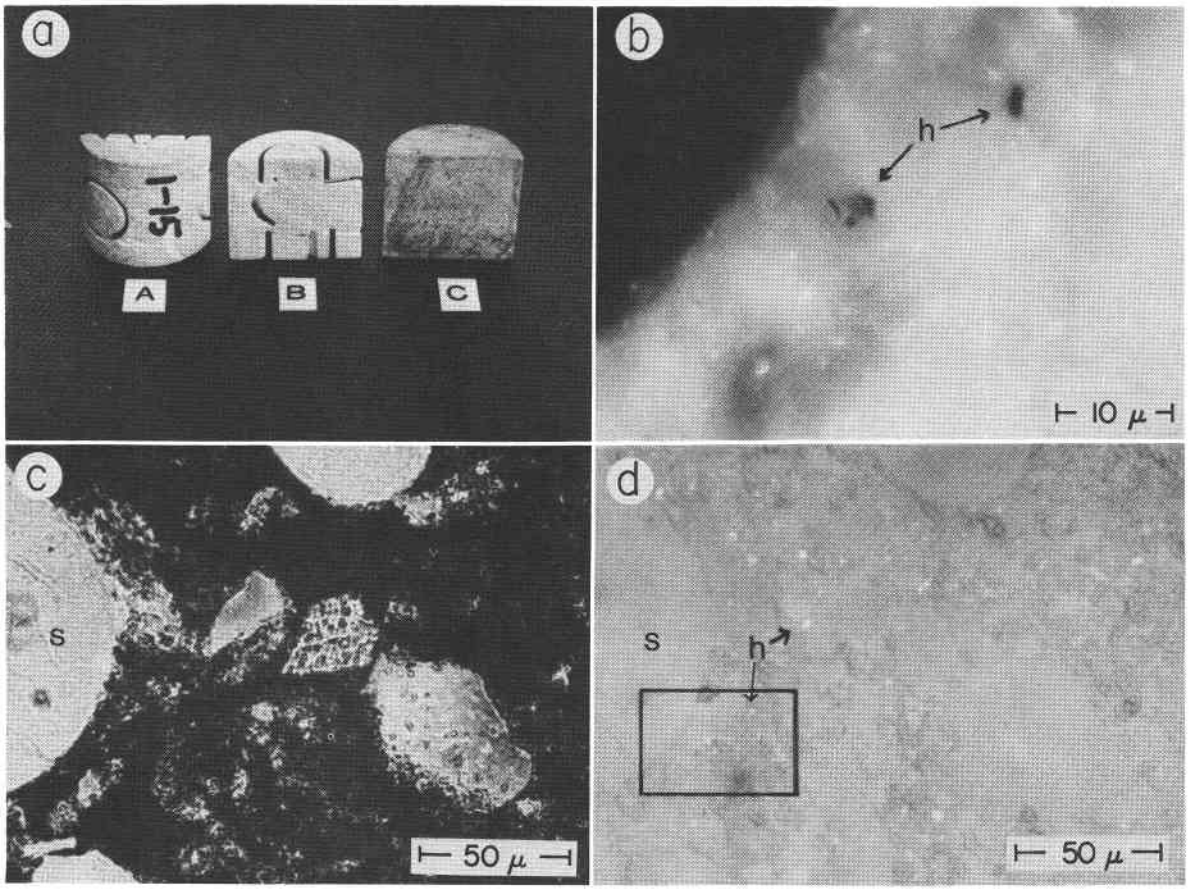


FIG. 5. (a) Photograph of 2.5 cm right cylinders of sandstone from Mauritania cut vertically; A and B after acid leaching showing the thin walled diamond drill cuts to allow acid penetration and the light grey colour after the red pigment has been dissolved out of the cement; C has not been leached in acid and is a typical red sandstone colour. (c, d) Photomicrographs at moderate magnification of polished thin section of sample 2-41 under transmitted light and reflected light respectively; S are sand grains, h are specular hematite grains; dark areas in (c) are the hematite pigmented cement. (b) A high magnification photomicrograph of the area in (d) outlined in black, taken with crossed polaroids so the hematite grains and also the sand grain show black.

and chemical cleaning revealed that the northerly directed and moderately inclined NRM of these specimens was composed of two separable magnetizations: an early, possibly original, I remanence whose weighted dip corrected mean direction is $D = 034°$, $I = +61°$, $\alpha_{95} = 6°$, and a later chemical remanence with a weighted dip corrected mean direction at $D = 336°$, $I = +37°$, $\alpha_{95} = 11°$ (Table 2). For both directions tilt correction produced an increase of Fisher precision (k); though not constituting a rigorous fold test, this favours a pre-folding magnetization.

Evidence for the relative age of the two magnetizations is based on their thermal and chemical treatment characteristics. Thermal treatment at progressively higher temperatures up to 400°C produced little change from the S direction. On further thermal treatment, the I direction was found either by direct measurement or by vector subtraction. Occasionally remanence domination by the late phase S magnetization totally suppressed all hints of the early phase I direction. The blocking temperature spectra of the two remanence carrying phases are overlapping so that for some samples, 4-41 in Fig. 4 for example, the S component has the higher Curie point, whereas for others, such as 3-21 in Fig. 4, the higher Curie point component defines the I direction. In this case thermal treatment alone cannot distinguish between initial and secondary magnetizations. Prolonged acid leaching distinguishes the components, however, by progressively removing the late phase S remanence, and revealing the early phase I direction.

In detail, the direction and intensity changes

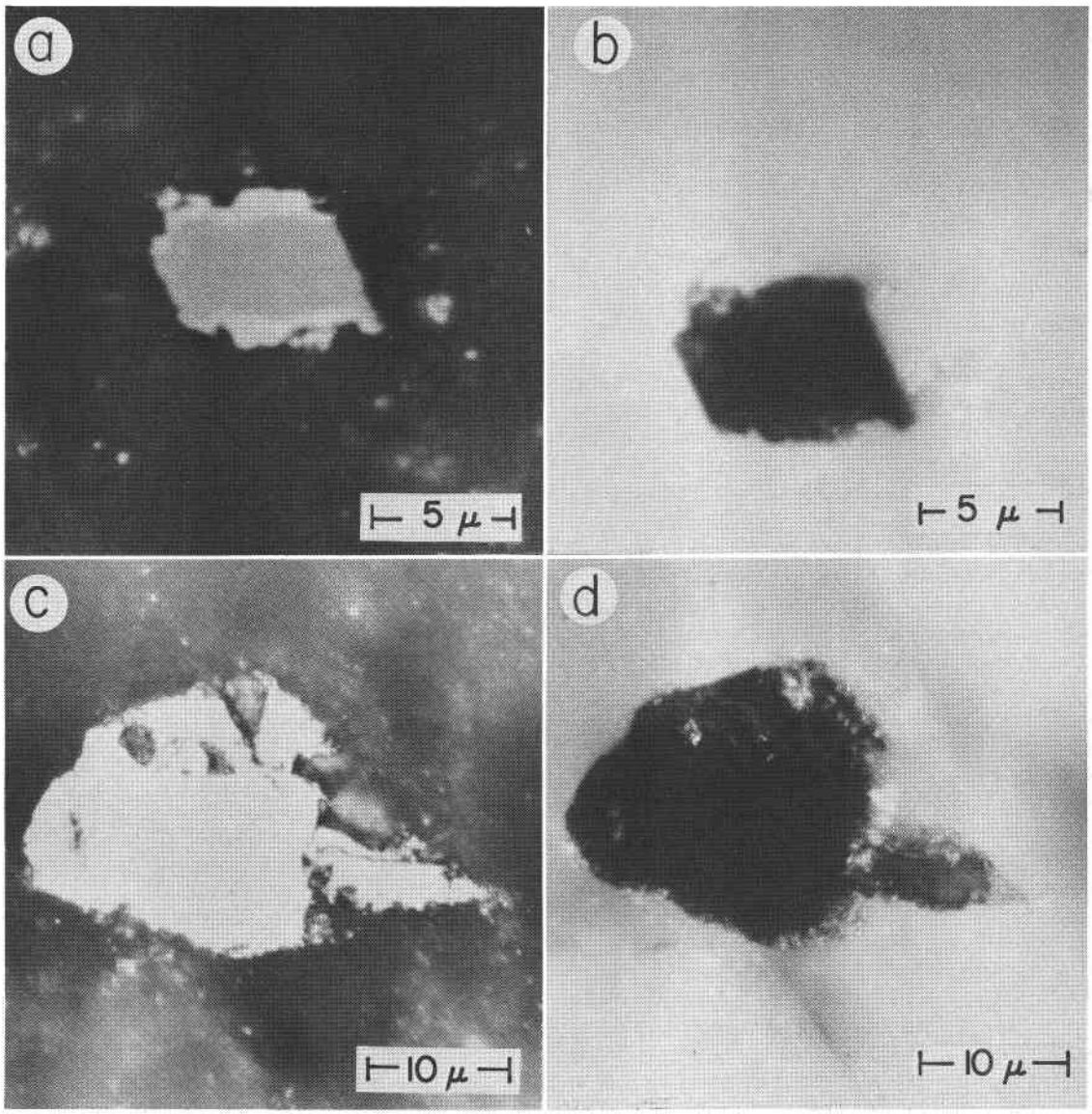


FIG. 6. Photomicrographs of a polished thin section from sample 6-41, showing (a) a specular hematite grain with white amorphous hematite overgrowth in reflected light; and (b) in transmitted light. (c, d) A larger hematite grain having amorphous hematite again on its outer edge, in reflected and transmitted light respectively. On both grains the amorphous hematite shows ruby red transluscence in reflected light.

produced by chemical leaching suggest the presence of three remanence carrying phases. As shown in Fig. 4, up to 585 h the successively revealed remanence directions fall approximately on a great circle path, associated with a very rapid decrease of remanent intensity. Between 585 and 1500 h the direction changes are mainly in inclination; the intensity decreases only slightly. After 1500 h of acid immersion further leaching slowly decreased the remanent intensity without further direction change. Vector analysis of these intensity and direction changes show the two 'S' phases removed have anti-parallel remanence directions, and hence were probably acquired sequentially by a similar type of hematite depositing process. The earliest I phase carried a significantly different direction and is probably related to a very different type of mineral phase. Figure 6, in which the three phases are shown to be (1) detrital hematite; (2) secondary overgrowth hematite on detrital grain; and (3) an omnipresent red hematite pigment explains their differing physical characteristics.

Leaching of the large amount of red hematite pigment from the sand to siltstone matrix is very quick, hence the rapid direction and large intensity changes. The overgrowth hematite, which is present in only small quantities, effects little intensity change, and is leached at a much slower rate because individual grains have to be attacked. The original detrital grains are then eventually open to degradation.

Discussion

All the presently available and relevant African and Gondwanic data, recently summarized by McElhinny and Embleton (1976) and Piper (1976), are plotted in Fig. 7 for the period 1100–650 Ma, and Fig. 8 for the period Cambrian to Ordovician. As the data are of highly variable paleomagnetic and (or) chronologic accuracy, some form of selection criterion must be applied in order to separate key results from other less reliable data. In this paper the classification system of McElhinny and Embleton (1976) has been adopted. Although not ideal, it adequately isolates the key paleomagnetic pole positions for the data discussed in this paper. Where a data set is more plentiful, more stringent selection criteria can be applied (Stewart and Irving 1974). Poles from the other Gondwanic continents are plotted after rotation to conform with the Smith and Hallam (1970) continental reconstruction.

In Fig. 7 the early phase magnetization (I_2-I) is not significantly different from other poles from south and east Africa, dated at approximately 1100 Ma. As this age is significantly different from the date reported for the Groupe de Char at 1019 ± 32 Ma, the question arises as to which magnetization this Rb/Sr date should be assigned. There are three possibilities: (A) the age is applicable to the I_2-I magnetization and there was little or no polar shift between 1100 and 1000 Ma. (B) the age is applicable to the I_2-I magnetization, but the late Precambrian polar wander for Africa is more complex than hitherto supposed. (C) the Rb/Sr age is applicable to the I_2-S magnetization, and closure of the isotopic system coincided with deposition of the red cement. I_2-I, meanwhile, should be assigned an 1100 Ma age.

The present evidence, though sparse, suggests the second possibility may be valid. Firstly, the Rb/Sr age is thought to record the final depositional stages of I_2 (Clauer 1973), so pole I_2-I should be temporally distinct from 1100 Ma poles. Secondly, the ages assigned to the other poles of Fig. 7 are not in temporal sequence. This problem could be overcome by a more complex apparent polar path for

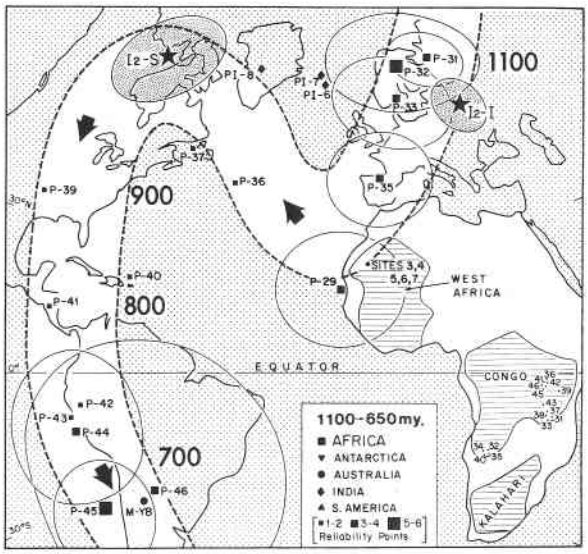


FIG. 7. A path of apparent polar wander relative to Africa for the interval 1100–650 Ma ago. Uncertainties associated with paleopole determination are emphasized by the 20 swathe width. Poles I_2-I and I_2-S are the initial and secondary poles from Formation I_2 of the Char Groupe. Poles from the Gondwanic continents, which are plotted after rotation to conform with the Smith and Hallam (1970) reconstruction, are identified by the following format; P-00 (e.g. P-34) are listed by Piper (1976) under individual continent listings, M-XX (e.g. M-YB) are listed by McElhinny and Embleton (1976), 0-000 (e.g. 2-101) are from the pole list of Irving and Hastie (1975). Symbols are plotted in three sizes, depending upon the total reliability classification of McElhinny and Embleton (1976). Confidence ovals (95%) are only plotted for those African results with three or more reliability points.

the period 1100–700 Ma. With the limited number of poles available, together with their poor age control, any such path would be highly speculative.

During the Cambrian to Ordovician period the Gondwanic poles define a polar loop (Fig. 8). Results from Africa for this interval are limited, but even so, the early magnetization of CO_{10}-I is statistically identical to lower Paleozoic results from Zambia (P-54), Morocco (P-56) and Algeria (2-101). Many of the pole positions for this time interval are poorly isolated; for example, only one of the poles from South America has an A_{95} less than 20° (Thompson 1973), and as seen from this collection incomplete chemical leaching of the red hematite pigment of the CO_8 specimens is reflected by the large confidence oval of the pole CO_8-I (Fig. 8). The chemical treatment does suggest that had better isolation been possible, a pole identical to CO_{10}-I would probably have been found.

A similar argument, however, cannot explain the apparently discrepant CO_8, CO_{10}-S pole, since its direction was apparently well isolated. Without in-

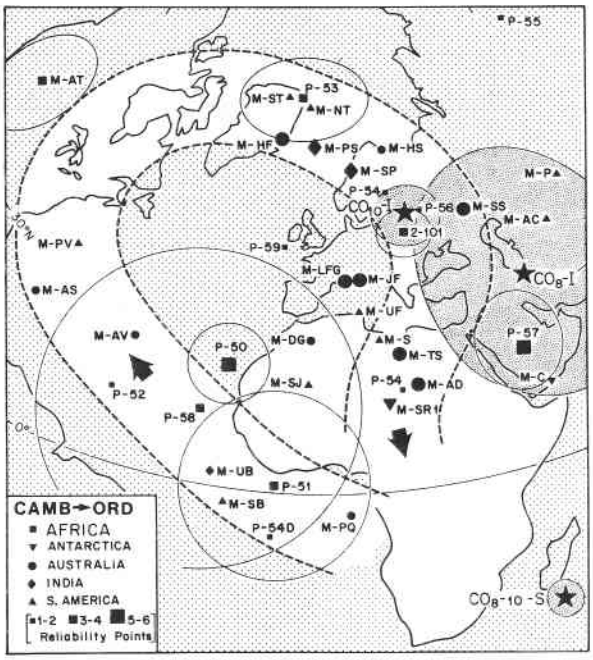


FIG. 8. A path of apparent polar wander relative to Africa for the interval late Precambrian to Ordovician. Uncertainties associated with paleopole determination are emphasized by the 20° swathe width. Poles CO_8-I and CO_{10}-I are the initial magnetizations of respectively formation CO_8 and CO_{10}, whereas pole CO_8-10-S represents the secondary remanence, which is identical at the two sites. Pole identification accords with that of Fig. 5, as does rotation of the Gondwanic poles to the Smith–Hallam (1970) fit. Symbols are plotted in three sizes, categorized by the reliability criteria of McElhinny and Embleton (1976). Confidence ovals (95%) are only plotted for those African results with three or more reliability points.

ternal field criteria (fold or contact test) assignment of this result to a particular time is dependent upon comparison with published results. A number of interpretations are possible, Permo–Carboniferous remagnetization perhaps being the simplest (Creer 1968; Schmidt and Morris 1977). Other more complex interpretations in terms of anomalous pole positions during the lower Palaeozoic (Thomas and Briden 1976) must await clarification of the detailed lower Paleozoic polar path for Gondwana.

In the absence of metamorphic, tectonic and deep burial effects, the other major factor that could effect secondary chemical alterations (remagnetizations) are probably climate dependent. Both the late phase magnetizations recorded by these studies were developed within 20° of the equator, hence it is possible that these two late phase magnetizations could record a period of lateritization, a process which is commonly observed in many present day low latitude formations (Schmidt and Embleton 1976).

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